CA1 - Specification

Individual Project

Weighting: **50%**

Due Date: **Sunday 15th November 2026, 23:55**

Objectives

- Obtain and load data from relevant sources
- To prepare the data for building a model which can classify between 24 different classes
- To explore and summarise the data
- To build a convolutional neural network model which can take in images and classify them between twenty four different classes
- To assess the accuracy of your model and minimise overfitting
- To use a source code repository (Git) to track the development of your solution

Learning Outcomes Addressed

- MLO3 Design and apply convolutional neural networks to identify between classes
- MLO4 Design and apply deep neural networks for fit complex datasets

Real-World Context

In this project, you are developing the image classification element of a self-driving car. You need to build a model which will identify between 24 different classes. You are required to build a convolutional neural network.

Assignment Brief

The CIFAR-10 and CIFAR-100 datasets can be downloaded from https://cave.cs.toronto.edu/kriz/cifar.html. The CIFAR-10 dataset consists of 60000 32x32 colour images in 10 classes with 6000 images per class, there are 50000 training images and 10000 test images. The dataset is divided into five training batches and one test batch, each with 10000 images. The test batch contains exactly 1000 randomly-selected images from each class. The training batches contain the remaining images in random order, but some training batches may contain more images from one class than another. Between them, the training batches contain exactly 5000 images from each class. You will only need the data for automobile, bird, cat, deer, dog, horse, and truck from this dataset.

The CIFAR-100 dataset is just like the CIFAR-10, except it has 100 classes containing 600 images each. There are 500 training images and 100 testing images per class. The 100 classes in the CIFAR-100 are grouped into 20 superclasses. Each image comes with a "fine" label (the class to which it belongs) and a "coarse" label (the superclass to which it belongs). You will only need the data for cattle, fox, baby, boy, girl, man, woman, rabbit, squirrel, trees (superclass), bicycle, bus, motorcycle, pickup truck, train, lawn-mower and tractor from this dataset.

Your objective is to build a convolutional model that can classify between automobile, bird, cat, deer, dog, horse, truck, cattle, fox, baby, boy, girl, man, woman, rabbit, squirrel, trees, bicycle, bus, motorcycle, pickup truck, train, lawn-mower and tractor as accurately as possible.

Screencast:

In the screencast you should:

- Detail and demo data pre-processing
- Demo and discuss data exploration
- Demo and discuss building the model
- Discuss overfitting and underfitting
- Detail testing the model

Version Control:

You are required to use version control to track the development of this project. You should read this article https://chris.beams.io/posts/git-commit/ and adhere to the seven rules detailed when writing commit messages. You should also commit regularly throughout the development of the project and add me as a collaborator on the project. You do not need to push the images and labels up to git and should adjust the .gitignore so that you don't as these files are big and you will overrun your allocated space.

Make sure that you name your repo with this naming convention st_ca1_firstname_last_name e.g. st_ca1_john_loane

Submission Checklist & Instructions

[Please submit a single .zip file on Moodle. When unzipped, your file must extract to a folder named st_ca1_firstname_lastname e.g. st_ca1_john_loane.

Critical	**CA Cover Sheet** Must be included and fully filled out.
Critical	**Code Repository** A link to your accessible repository in a Readme.md file. Ensure you have shared the repo with your lecturer. All work must be committed to version control following the Chris Beams commit message rules. Do not commit the image data.
Critical	**Screencast / Demo** A screencast demonstrating the development and testing of your model
Required	**Source Code** Source code, along with link to repo and screencast submitted in one .zip on Moodle
Critical	**AI Declaration & Log**

	An AI_Chat_Log.md file in your repo and a completed AI usage cover sheet.

(**Note:** "Critical" elements mean your work will not be marked without them.)

Marking Criteria

Primary Category	Sub-Criteria	Weight	Description
Data Pre-Processing	Loading in the data from two different sources Cleaning Combining	30%	The data is not perfectly set up for your purposes in building a model to classify objects. You will need to load in the data from CIFAR-10 and CIRFAR-100, extract the relevant data from each dataset and combine them together. You may also need to preprocess the data to scale, grayscale, adjust lighting, and scale down variability.
Data Exploration	How many images? Image size? Classes?	10%	Having pre-processed the data you should investigate the images that you have for each class. What size are the images? What classes do the images represent? You can also present any work you have done before processing to investigate the data such as counting how many labels each image has.
Building the model	Models built Overfitting Underfitting	30%	Build an appropriate model to classify between the 24 different classes. Consider **overfitting** and **underfitting** when building the model. Comment on these issues and how you have addressed them in the report.
Testing the model	Accuracy Hyperparameters Loss Regularisation	10%	Use the test data to measure the accuracy of the model. This should be an iterative process where you tweak the hyperparameters to improve the accuracy of the model between test runs. Document your results and the hyperparameters changed between each run and how the accuracy has improved.
Screencast	Demonstration and discussion	10%	In the screencast you should: • Detail and demo data pre-processing

			• Demo and discuss data exploration • Demo and discuss building the model • Discuss overfitting and underfitting • Detail testing the model
Version Control	Version Control & Repository Management	10%	You are required to use version control to track the development of this project. You should read this article https://chris.beams.io/posts/git-commit/ and adhere to the seven rules detailed when writing commit messages. You should also commit regularly throughout the development of the project and add me as a collaborator on the project. You do not need to push the images and labels up to git and should adjust the .gitignore so that you don't as these files are big and you will overrun your allocated space.
In-class verification	Written class test based on your submission	0/1 If you do not pass this your work will get a mark of zero	10 custom questions for each student based on your submission. You will be required to answer these questions with pen and paper in class after the submission deadline. If your answers to these questions show no understanding of the work a multiplier of 0 will be applied to your submission.

Method of feedback to be provided to students:

Verbal

Generative AI Policy

Extent of AI use[1]:

[1] Consult the Generative Artificial Intelligence Guidance for Staff for narrative to be included here.

Open Use

Generative artificial intelligence (AI) tools are not restricted for this assessment task. In this assessment, you can use generative artificial intelligence (AI) to assist you in any way.

Any use of generative AI must be appropriately acknowledged (in accordance with DkIT Academic Integrity Policy and Procedures, https://www.dkit.ie/about-dkit/policies-and-guidelines/academic-policies.html).

Notes to Students

- Please note that all work submitted as partial fulfilment of this module must adhere to the Institute Academic Integrity Policy.
- If generative artificial intelligence (AI) is permitted in the assessment and the learner has chosen not to use it, the following disclosure in the learner submission is recommended: 'No content generated by AI technologies has been used in this assessment'.
- We recommend all students consult the DkIT Guide to Harvard Referencing which provides useful information on referencing and how to avoid plagiarism: https://www.dkit.ie/documents/dkit-referencing-guidelines-and-how-avoid-plagiarism
- All work must be submitted by the due date indicated by your lecturer(s). Any work submitted after the due date is subject to the penalties outlined in the Institute Continuous Assessment Procedures.